

Mid-term I Adv Quantum Mechanics

Phys 336 OQS problems

April 29th 2026

Honor code: This is a take-home test run on the honor system, see below.

Due date: Solutions are due back to me Wed May 6th by 3PM – you can hand it to me or in the HW mailbox opposite Trenne’s office.

Open notes: You may ONLY use your notes and HW, and anything I have provided in the ‘SharedNotes+’ directory. I have linked to other material on moodle – don’t use that! You can ask me for help as well.

Using symbolic or numerical packages: You are welcome to use previously coded up symbolic manipulation packages or numerical packages to work on the problems below. It’s okay if it is adapted from stuff you OR ANYONE ELSE in the class previous wrote up but you should already have access to it. Always cite the source in our solutions.

Time: There is a 5 point incentive for finishing the written part of the test in under 3 and a half hours. You do not have to spend the time continuously. For example, bio breaks, including for eating and sleeping are perfectly okay. However, please understand that opening the test to look at the problems and then shutting down to think about them or looking at notes before starting, or walking away to think about the problem etc would amount to violation of the point of this; time spent on the test is not ONLY the writing time, in short. Please record your times for every start and stop, and note the total time as well.

Record on the submitted test: On the test, please record your time information as well as the honor pledge “On my honor, I have neither given nor received any unauthorized help on this test”. Sign below this written pledge.

READ THIS CAREFULLY before starting:

Do 4 problems total, choosing one from each of 1st 3 sections’ below; I have tried to construct them so they are about equally difficult within the section. I will also weigh all three sections equally. Your 4th ‘problem’ is to schedule a 20 minute discussion with me.

Measurement: 25 points

Problem 1A: The Trine POVM and state discrimination

In a $d = 2$ Hilbert space, consider three non-orthogonal states separated by 120° in the xz -plane, where the task is to increase the probability of correctly identifying one of the states presented to you at random (with equal probability) while minimizing errors. This is a VARIATION on what we had done earlier. The three states are

$$|\psi_0\rangle = |0\rangle, \quad |\psi_1\rangle = -\frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle, \quad |\psi_2\rangle = -\frac{1}{2}|0\rangle - \frac{\sqrt{3}}{2}|1\rangle$$

- (a) Show that the operators $E_k = \frac{2}{3}|\psi_k\rangle\langle\psi_k|$ for $k \in \{0, 1, 2\}$ form a valid POVM.
- (b) If you use this POVM, what is the probability of correctly identifying $|\psi_0\rangle$ if you are indeed given $|\psi_0\rangle$ and asked to identify it? What is the probability of erroneously thinking it is one of the other states? How do these two numbers change for $|\psi_1\rangle, |\psi_2\rangle$?
- (c) If the ‘success probability’ P_{succ} is the probability of successfully identifying the state as one of $|\psi_0\rangle, |\psi_1\rangle, |\psi_2\rangle$ correctly given one of the three $\{|\psi_k\rangle\}$ with equal prior probability $1/3$, calculate a value for P_{succ} .
- (d) How would this success probability have worked if you had used any pair of projective measurements via orthonormal basis vectors (feel free to use one or more explicit bases to calculate this)? Which works ‘better’ (gives a higher success probability)? What does this tell you about POVMs?

Problem 1B: POVMs/Kraus non-uniqueness.

Consider an amplitude-damping channel for the 2-state atom (at zero temperature) that might decay releasing a photon into an electromagnetic field; below we work out how this channel is related to a POVM. In this case, say that the atom is initially in the state $|\psi_0\rangle = \frac{1}{\sqrt{2}}(|1\rangle + |0\rangle)$.

- (a) Show that the operators M_0, M_1 satisfy the constraint $\sum_\alpha M_\alpha^\dagger M_\alpha = I$ where

$$M_0 = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{p} \end{pmatrix} \tag{1}$$

$$M_1 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}. \tag{2}$$

- (b) Given the initial $|\psi_0\rangle$ as above, what are the probabilities of the two measurement outcomes M_0 and M_1 and what are the final states after each outcome respectively?
- (c) Show that the Kraus operators M_2, M_3 defined below also satisfy $\sum_\alpha M_\alpha^\dagger M_\alpha = I$

$$M_2 = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix} \tag{3}$$

$$M_3 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}. \tag{4}$$

- (d) Again given $|\psi_0\rangle$ as above, what are the probabilities of the two measurement outcomes M_2 and M_3 and what are the final states after each outcome respectively ?
- (e) Show that $\{M_2, M_3\}$ measurements correspond to the **same** POVM as for $\{M_0, M_1\}$.
- (f) What is the unitary matrix transforming M_0 to M_2 ? (If you prefer you can find the one that transforms M_2 to M_0). Use this to comment on the differences and similarities between the two sets of measurement actions in (b) and (d) above and try to connect it to the physics interpretations for this atom and field situation.
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OSRs: 25 points

Problem 2A: Depolarizing channel

Consider an error channel that either takes a qubit to the maximally mixed state $I/2$ with probability p ('erases the polarization') or leaves it alone with probability $1 - p$ so that we can write

$$\rho' = pI/2 + (1 - p)\rho, \quad (5)$$

which is, however, not in the Kraus OSR form because of the I term.

- (a) To put it into the Kraus OSR form, first show that for any qubit density matrix,

$$\rho + \sigma_x \rho \sigma_x^\dagger + \sigma_y \rho \sigma_y^\dagger + \sigma_z \rho \sigma_z^\dagger = 2I \quad (6)$$

- (b) Use Eq. (2) to replace the troubling term in Eq.(1) for ρ' and make it into Kraus OSR form; What are the resulting Kraus operators ? Verify that they satisfy $\sum_\alpha K_\alpha^\dagger K_\alpha = I$.
- (c) Writing ρ in the Bloch vector expansion form $\rho = \frac{1}{2}(I + \vec{v} \cdot \vec{\sigma})$ describe how the Bloch vector $\vec{v}$ transforms under this channel (it might be easier to start from Eq. (1) above for this, but get the answer any way you like).
- (d) Recalling that all Kraus OSRs on the Bloch sphere correspond to an affine transformation of $\vec{v}$, what is the form of the transformation matrix M ?

How would you describe the general action of this particular quantum error channel on the Bloch sphere ?

Problem 2B: Qutrit Dephasing OSR

For a qutrit ($d = 3$), define the generalized Pauli Z operator such that $Z|n\rangle = \omega^n|n\rangle$, where $\omega = e^{2\pi i/3}$ (it's the same Z operators as last time, just written differently). Consider a channel Φ defined by the Kraus operators:

$$M_0 = \sqrt{1-p}I, \quad M_1 = \sqrt{\frac{p}{3}}I, \quad M_2 = \sqrt{\frac{p}{3}}Z, \quad M_3 = \sqrt{\frac{p}{3}}Z^2$$

- (a) Verify that this set satisfies the completeness relation for $p \in [0, 1]$.
- (b) Determine the effect of this channel on the off-diagonal elements ρ_{01} and ρ_{02} of a general qutrit density matrix.
- (c) Identify the stationary states of this channel.
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Applied problems: 25 points

Problem 3A: The Generalized Amplitude Damping Channel (OSR)

A qubit is coupled to a thermal bath at temperature T . The Operator Sum Representation (OSR) is given by:

$$\begin{aligned} E_0 &= \sqrt{p} \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, & E_1 &= \sqrt{p} \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix} \\ E_2 &= \sqrt{1-p} \begin{pmatrix} \sqrt{1-\gamma} & 0 \\ 0 & 1 \end{pmatrix}, & E_3 &= \sqrt{1-p} \begin{pmatrix} 0 & 0 \\ \sqrt{\gamma} & 0 \end{pmatrix} \end{aligned}$$

- (a) Find the stationary state ρ_∞ such that $\Phi(\rho_\infty) = \rho_\infty$, i.e. acting on it with the full map does not change ρ .
- (b) Use this to relate p to the temperature T using the Boltzmann distribution where the upper state has exponentially smaller population as

$$\frac{\rho_{11}}{\rho_{00}} = e^{-\hbar\omega/k_B T}$$

where $\hbar\omega$ is the energy difference between the two qubit levels.

Problem 3B: Driven-dissipative qubit (Master equation)

A qubit is driven by a resonant laser (Rabi frequency Ω) and undergoes spontaneous emission at rate γ . The Master Equation in the rotating frame is:

$$\dot{\rho} = -i\left[\frac{\Omega}{2}\sigma_x, \rho\right] + \gamma\left(\sigma_-\rho\sigma_+ - \frac{1}{2}\{\sigma_+\sigma_-, \rho\}\right)$$

- (a) Write the Bloch equations for $\langle\dot{\sigma}_x\rangle, \langle\dot{\sigma}_y\rangle, \langle\dot{\sigma}_z\rangle$.
- (b) Find the steady-state population inversion $\langle\sigma_z\rangle_{ss}$ as a function of the saturation parameter $s = 2\Omega^2/\gamma^2$.

Discussion: 20 points

Problem 4: Meeting

Please schedule a 20 minute meeting with Arjendu, using slots available on his calendar. These could be by zoom in the evening as well, if no daytime slots are available. This should be scheduled for at least 24 hours AFTER you have submitted the test (or planning to submit the test, we can move it if things change). In preparation for the conversation, you should gather your ‘pedagogical problem’ submissions in one place, and read over the test itself again.